

# Covariance & Correlation

Covariance and Correlation are two statistical measures used to determine the relationship between two variables. Both are used to understand how changes in one variable are associated with changes in another variable.

Covariance :-

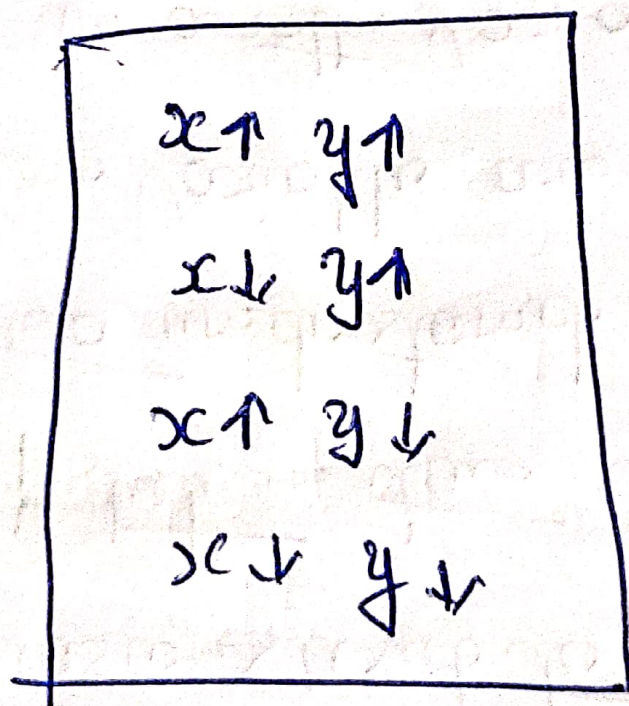
Covariance is a Measure of how much two random variables change together. If the variables tend to increase



and decrease together, the covariance is positive. If one tends to increase when the other decreases the covariance is negative.

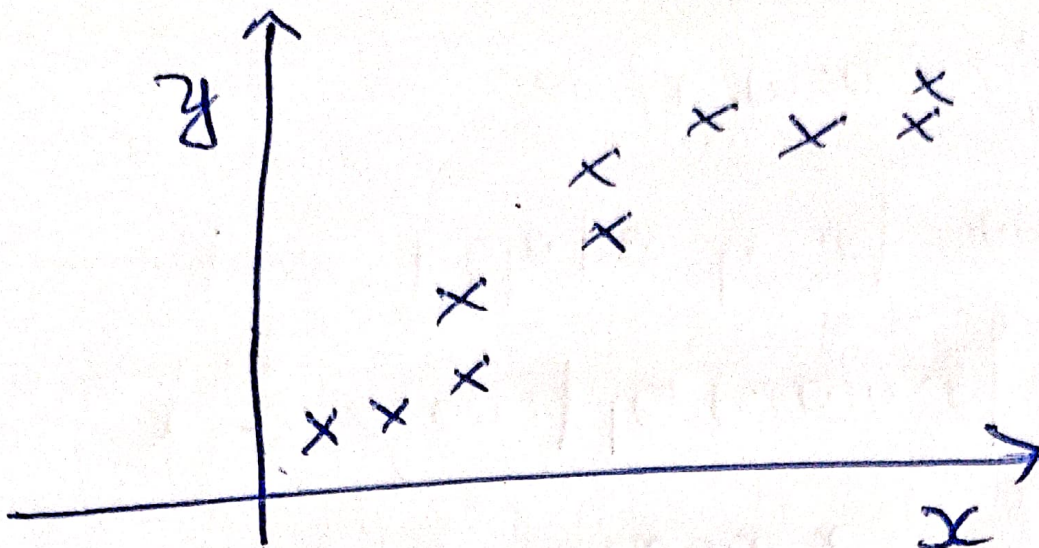
Quantify the Relationship b/w  $x$  &  $y$

|   | $x$ | $y$ |
|---|-----|-----|
| → | 2   | 3   |
| → | 4   | 5   |
| → | 6   | 7   |
| → | 8   | 9   |



| Size of House | Price    |
|---------------|----------|
| 1200          | 45 lakhs |
| 1300          | 50 lakhs |
| 1500          | 75 lakhs |





|                |                |
|----------------|----------------|
| $x \uparrow$   | $y \uparrow$   |
| $x \downarrow$ | $y \downarrow$ |

$\Downarrow$

+ve correlation  
+ve value



| $x$   | $y$ |
|-------|-----|
| 7     | 10  |
| 5     | 12  |
| <hr/> |     |
| 3     | 14  |
| 4     | 16  |

|                |                |
|----------------|----------------|
| $x \downarrow$ | $y \uparrow$   |
| $x \uparrow$   | $y \downarrow$ |

-ve correlation  
-ve value.



# Covariance

$$\text{Cov}(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\text{Cov}(x, x) = \sum_{i=1}^n \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$$

$$\boxed{\text{Cov}(x, x) = \text{Var}(x)}$$

$$\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

$x_i \rightarrow$  Data point of random variable  $x$

$\bar{x} \rightarrow$  Sample mean of  $x$

$y_i \rightarrow$  Data points of random variable  $y$

$\bar{y} \rightarrow$  Sample mean of  $y$



Students

Hours Studied (x)

Exam Score (y)

2

50

3

60

4

70

5

80

6

90

$x \uparrow \quad y \uparrow \quad \quad x \downarrow \quad y \downarrow$   
+ve Covariance

$$\textcircled{1} \bar{x} = \frac{2+3+4+5+6}{5} = \frac{20}{5} = 4$$

$$\textcircled{2} \bar{y} = \frac{50+60+70+80+90}{5} = \frac{350}{5} = 70$$

$$\begin{aligned} \text{Cov}(x, y) &= (2-4)(50-70) + (3-4)(60-70) \\ &\quad + (4-4)(70-70) + (5-4)(80-70) \\ &\quad + (6-4)(90-70) \\ &= 4 \end{aligned}$$



$$= \frac{(-2)(-20) + (-1)(-10) + (1)(10) + 2(20)}{4}$$

$$= \frac{100}{4} = 25$$

$\rho_{xy} = 25$  +ve covariance.

No. of hours in exam scores.

| x | y  |
|---|----|
| 2 | 10 |
| 6 | 12 |
| 5 | 14 |

Advantages

① Quantify the Relationship  
w/w x only

Disadvantages

① Covariance does not have a specific limit value.

$\rho_{xy} \rightarrow \infty$  to an

② Correlation  $\begin{cases} \rightarrow \text{Pearson Correlation Coeff} \\ \rightarrow \text{Spearman Rank Correlation} \end{cases}$

1) Pearson Correlation Coefficient  $\Rightarrow [-1 \text{ to } 1]$

$$\rho_{x,y} = \frac{\text{Cov}(x,y)}{\sigma_x \cdot \sigma_y}$$

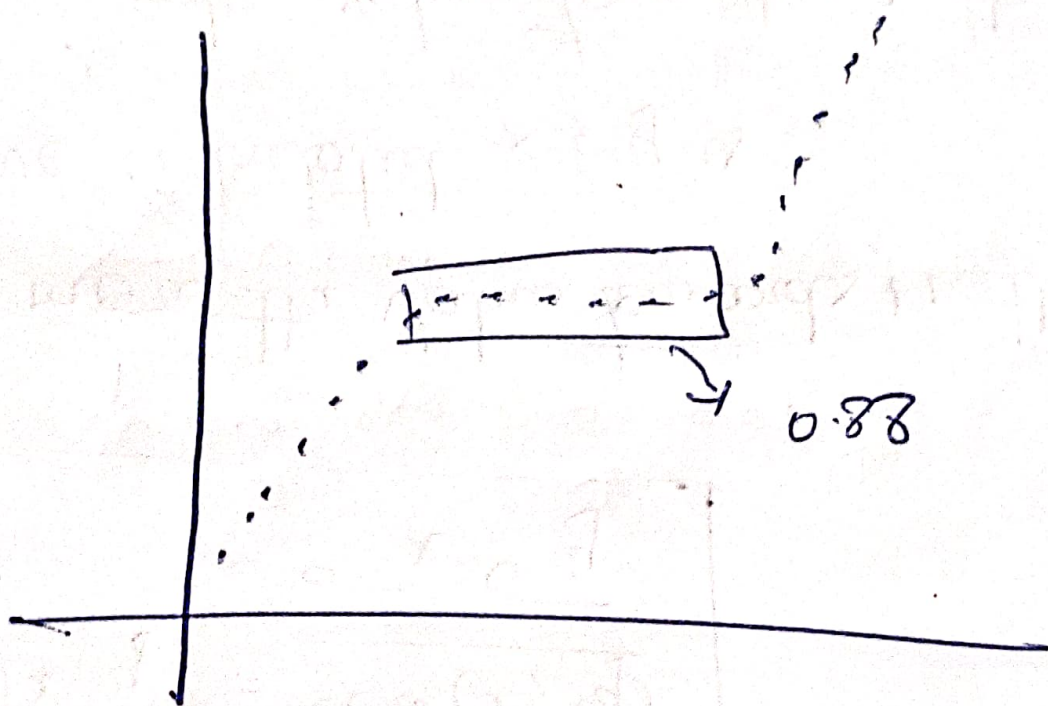
⊛ The more the value towards  $+1$  the more +ve correlated  $x$  &  $y$  is.

⊛ The more the value towards  $-1$  the more -ve correlated  $x$  &  $y$  is.



$$\frac{20}{\sigma_x \cdot \sigma_y} \Rightarrow \boxed{-1 \text{ to } 1}$$

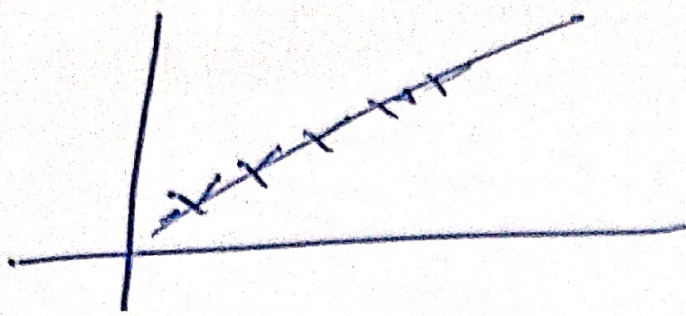
ii) Spearman Rank Correlation



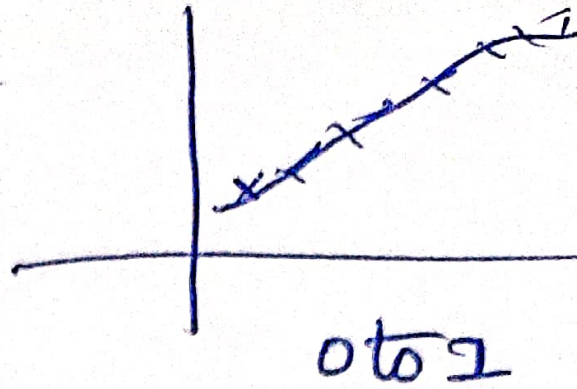
$$\text{Spearman} = 1$$

$$\text{Pearson} = 0.88$$





$\rightarrow 1$



0 to 1

Pearson Correlation is unable to capture  
Correl for non linear data



$$\rho_s = \text{Corr}(R(x), R(y))$$

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$$\sigma(R(x)) * \sigma(R(y))$$



| x | y | R(x) | R(y) |
|---|---|------|------|
| 1 | 2 | 2    | 1    |
| 3 | 4 | 3    | 2    |
| 5 | 6 | 4    | 3    |
| 7 | 8 | 5    | 5    |
| 0 | 7 | 1    | 4    |

Feature Selection.

No. of p  
o/p

Size of Rows  $\uparrow$  No. of Rows  $\uparrow$  Location  $\uparrow$  Price  $\uparrow$